

Name-and-Birthdate Matching in Voter Purges

Two people who have never met, one match key, and a purge rule built on the assumption that such a thing cannot happen

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Somewhere in New Jersey there are two people who have never met. They live in different counties. They have different families, different addresses, and no connection to each other of any kind. They also have the same first name, the same last name, and the same date of birth – not the same birthday, the same *date*, year included.

There are 4,531 such people in that state's registration file, and the file cannot tell them apart. Neither can any program built to read it.

For roughly a decade, a consortium of states ran exactly such a program across their combined rolls, treated the matches it produced as evidence of duplicate registration, and used them to justify removing voters. The reasoning is easy to sympathize with. Two records, same name, same birth date: what are the odds?

That is the right question. It has an answer, and the answer is not the one intuition supplies.

01 Two states, and only one of them writes down the day

Neither file behind this chapter was published for it, and neither one leaves the folder it was read from.

The main source is New Jersey's registration list, which the state releases county by county. That is 21 separate extracts carrying an identical layout, which is what makes reassembling them a legitimate operation rather than a guess. It arrived here as working material in a redistricting matter, which is the ordinary route for a file of this kind. It is a public record either way. The second source is one county's registration extract from Georgia. Houston County, 127,560 records, the same file the BISG chapter grades a sur-name model against.

Two states is one more than the argument strictly needs, and the extra one is the whole design. **New Jersey publishes a complete date of birth. Georgia publishes a birth year.** The Interstate Crosscheck program matched on first name, last name and full date of birth. So New Jersey is the place where that rule can be *reproduced* rather than approximated: the actual key, applied to actual registrations, with the actual collisions counted.

Georgia is where the same idea has to run on a far coarser key. A birth year is all a state that withholds the day and the month leaves anyone to work with. Between them, the

precision of the match key becomes something this chapter can vary and measure, rather than something it has to argue about. Very few pairs of public files allow that.

New Jersey's counties are also separate jurisdictions inside one file. That makes a match spanning two of them a usable stand-in for the cross-state match Crosscheck existed to return. It is a stand-in, and a later section is careful about the difference.

What is being counted was settled once, and saying so matters, because the alternative would have flattered the file. **Every registration record is counted here, whatever its status.** 6,145,992 of New Jersey's 6,657,135 carry a status beginning "Active". The rest are inactive or in one of several conditional states, and they are counted too. List-maintenance programs act on the roll as a whole, and an inactive registration is exactly the kind most often put forward for removal. Restricting to active records would have measured a friendlier list than the one a purge is actually run against.

Cleaning, then, was almost all subtraction, and the subtraction happens early. A handful of columns from each county file are read, and the rest are discarded as the file is parsed. Only 4 records in the entire state carry no date of birth, so nothing needed imputing and nothing was imputed. To *impute* is to fill in a value nobody recorded, by guessing it from other things.

Nothing is fetched. Both extracts are read from local disk, with no network, no key and no guestbook. And nothing that leaves those files is a person. What is kept here is counts and distributions only: no name, no date, no registration number, no address. The raw extracts are never copied alongside them.

That is a statement about the *derived* files, which is where the risk of quiet reuse actually lives. A committed key file would be a purge list sitting in a teaching repository, indexed and copyable. Five keys printed in the body of an argument about purge lists are a different object, and the chapter prints them for the reason given below.

None of that is the same as reproducible, and the difference is worth stating plainly. There is no address in this chapter for its main source, because the file did not arrive by download at all. It arrived because somebody was already working with it.

Both states will hand these lists to anyone who asks, which is what makes them public records. But asking is a form, a fee and a wait rather than a line of code. And the 21 county extracts still have to be stitched back together at the far end of it.

So a reader has a slower road to the raw records than the person who wrote this did. The decision not to commit the extracts makes it slower still. The alternative was shipping 6,657,135 people's names and birth dates in a teaching folder. That is not a trade worth making for convenience.

02 The file anybody can have

It matters that this is not a secret document. New Jersey's registration list is a public record, and so is Georgia's; either state will hand a copy to anyone who fills in the form and pays the fee. **That is not a loophole in the argument of this chapter. It is the argument.** A matching rule anyone can run, against a list anyone can obtain, is a live capability rather than a hypothetical one. A chapter that showed you a scrubbed imitation of the file would be teaching you that it is harder to get than it is.

So these are real rows. A New Jersey county extract has 28 columns:

No codebook ships with the extract, so the third column is read off the names and the values rather than quoted from a document:

#	Column	What it holds	Read here
1	displayId	the identifier the state shows on screen	—
2	leg_id	a second, internal identifier	—
3	party	party affiliation, which New Jersey does register	—
4	status	active or inactive	read
5	reg_date	when the registration was made	—
6	last	surname	read
7	first	given name	read
8	middle	middle name	read
9	suffix	Jr, III, and so on	—
10	dob	full date of birth — day, month and year, unlike Georgia	read
11	street_num	house number	—
12	street_pre	the N in N Main St	—
13	street_post	the NW in Main St NW	—
14	street_base	the street's name without its type	—
15	street_suff	St, Ave, Rd	—
16	street_name	the assembled street name	—
17	apt_unit	apartment or unit	—
18	city	city	—
19	zip	ZIP	—
20	county	county	read
21	municipality	municipality	—
22	ward	ward	—
23	district	election district	—
24	congressional	U.S. House district	—
25	legislative	state legislative district	—
26	freeholder	county board district — New Jersey's old name for it	—
27	school	school district	—
28	fire	fire district	—

Six of them are read. Here are five registrations in those six, drawn at random from all 50,227 records in the Salem County extract — not the first five, which in an alphabetized

file would all be surnames beginning AA:

last	first	middle	dob	county	status
FOX	LESTER	H	1953-09-18	Salem	Active
HALSTEAD	TAMMY	L	1975-05-11	Salem	Active
MERCKX	VIRGINIA	M	1943-01-06	Salem	Active
SCARDUZIO	JACLYN	S	1978-04-24	Salem	Active
SZYMANSKI	JACOB	A	1994-03-30	Salem	Active

Every field is exactly as New Jersey publishes it, **including the full date of birth**, and that is deliberate. Name plus full date of birth is the Crosscheck key itself, so these five lines are five complete keys. That is the only honest way to show what the rule operates on. A version with the month and day starred out would be easier to print and would misrepresent the situation twice over. It would suggest the state withholds something it does not. And it would let a reader believe the key is harder to assemble than it is.

The discomfort of reading five real dates of birth is the correct response to a rule that ran against 6.6 million of them. Nothing here is disclosed that the state does not sell to anyone who asks. The chapter's argument is precisely that this is so.

The other 22 columns are never loaded at all. They are refused before the file is in memory. That is partly arithmetic: the 21 extracts are about 1.2 gigabytes of text, and the six columns are a few hundred megabytes. It is partly a matter of appetite.

The rule being examined needs a name and a birth date. It has no use for where anybody lives, so the parser never learns. A street address that is never read cannot leak from an analysis that never had it.

Two things in those five rows are worth more than they look. **county is the same value in every row**, because a county extract is one county. The column only starts carrying information after all 21 files are stacked, which is exactly when a match spanning two of them becomes findable. And **status is not a yes-or-no field**. Salem alone uses seven values: Active, Inactive Confirmation, Active Need ID, Pending-17, Pending, Inactive Confirmation Need ID, and Pending-17 Need ID. That is why what counts as a record here had to be settled deliberately, rather than by filtering on the word "Active".

Then everything the chapter is about happens in one line. The six columns become one string:

TAMMY|HALSTEAD|1975-05-11

That string is the entire matching rule. Uppercase it, trim it, join it with a separator, and a person has been reduced to a key. Sort 6,657,135 of those keys, count how often each one repeats, and you have measured the Interstate Crosscheck program.

What leaves the build script is not a list of keys. It is a count of how they repeated:

Stage	Rows	A row is
The 21 county extracts	6,657,135	one registration, 28 columns
One key per record	6,657,135	one registration, six columns and a key

Stage	Rows	A row is
Distinct keys	6,653,955	one name-and-birth-date, however many people hold it
Keys held by more than one registrant	3,149	one collision
What ships in data/keys.csv	6	one matching rule, with its collision counts

6,657,135 people's names and birth dates go in. A file with 6 rows and no names in it comes out.

The decisions

Every record counts, not just the active ones. 6,145,992 of the 6,657,135 registrations carry a status beginning "Active." The other 511,143 are counted too. Restricting to active registrations would have measured a tidier list than the one a purge is actually run against, and it is precisely the inactive registrations that purges go after.

Case and spacing are flattened before anything is compared. Names are uppercased and trimmed, so Pat and PAT become the same string. That step *creates* matches. Two records that differed only in a trailing space now collide. It is the right thing to do, because the difference was never about a person. It is still worth seeing that a cleaning step and a matching step are the same operation here.

The middle name becomes a single letter. Where the key uses it at all, it uses the initial. A file in which some people wrote *Q* and some wrote *Quentin* cannot be compared on the full field. Every such abbreviation makes the key coarser and the collisions more numerous, and those collisions are what the rest of this chapter counts.

Nothing that identifies a person is written out. The largest group of New Jersey registrants sharing a first name, a last name and a full date of birth has 4 members. Publishing the number 4 is safe. Publishing the name and date those 4 people share would be building the purge list this chapter exists to criticize, so what is kept here is counts and distributions and nothing else.

Nothing was imputed, and one thing was left conspicuously alone. Exactly 4 records in the whole state carry no date of birth, so there was nothing to fill in. But some of the dates that *are* present are not birth dates at all. They were left exactly as the state wrote them. This chapter comes back to that decision, because the result is one of the most revealing things in the file.

03 Twenty people in a room

The cleanest way to see how badly intuition performs here has nothing to do with voting.

Take a room of people and ask whether any two of them share a birthday – just the day, ignoring the year. There are 365 of those to go around. This class has 20 people in it, 19 students and an instructor, so the room is sparse: 20 people spread over 365 possible dates fills about 5% of them.

Before going further, commit to two numbers.

Before you read on... In a room of 20 people, what is the probability that at least two of them share a birthday? And how many people would you need in the room before that probability passes even odds – before a shared birthday is more likely than not? Write both down. Most people, asked the second question, answer somewhere between 100 and 200.

The answers are 41.1% and 23.

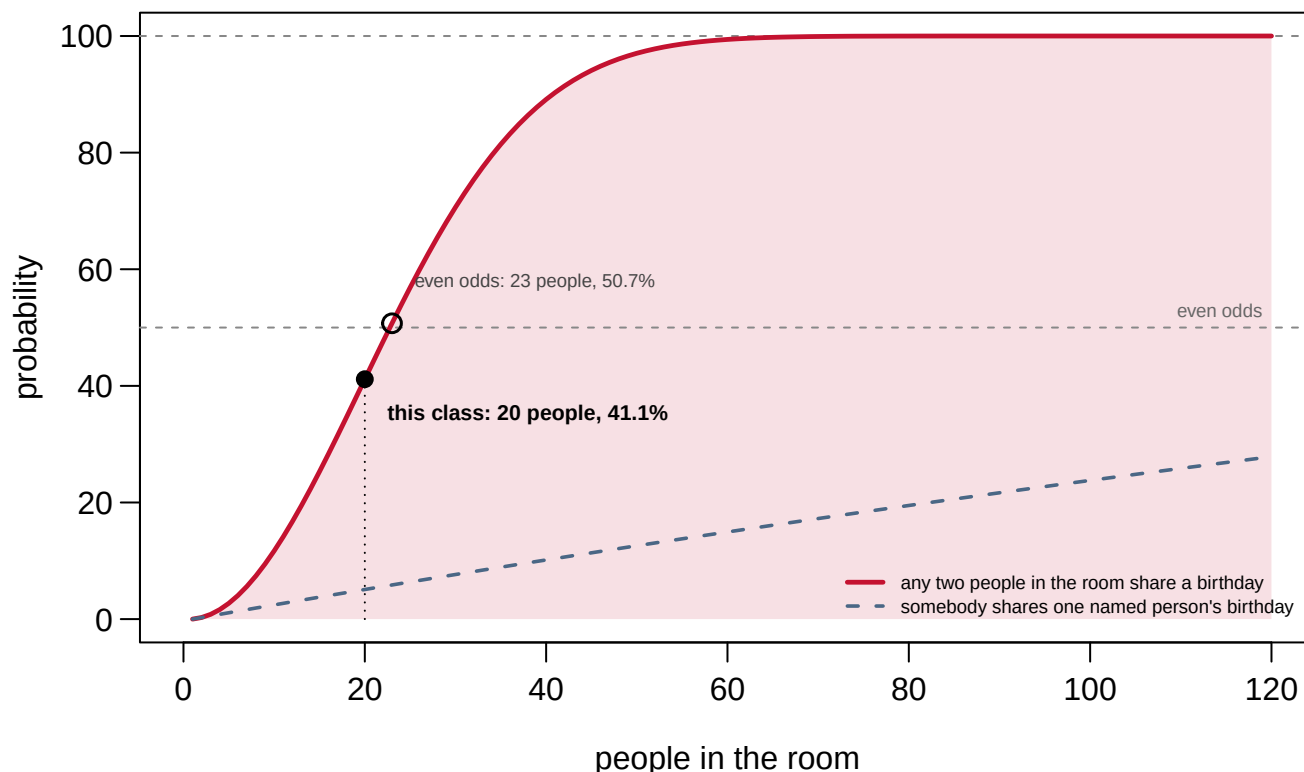


Figure 1. The chance that a room of any given size contains a shared birthday. Computed exactly, not by the usual approximation. At 20 people the probability is 41.1%; it passes even odds at 23 people and reaches 97% at 50. The dashed line is the question readers usually answer by mistake: the chance that somebody shares *one named person's* birthday, which at 20 people is only 5.1%. **In the browser version the box above the figure sets the size of the room**, so the class can be counted on the day; the printed version is fixed at 20. Percentages are rounded to one decimal, and a reading of >99.9% at the right-hand end is not certainty – see below.

23 people. Not 183. The guess of 183 feels right because it answers the dashed curve's question rather than the solid one. That question is how many people you need before somebody shares *your* birthday, which really does take about half a year's worth of people. But nobody asked about you.

And the curve never actually reaches certainty. Push the room size to 120 and the chance of a shared birthday is >99.9%. This chapter prints it that way rather than as 100.0%, because it is not 100%. A shared birthday is genuinely unavoidable only at 366 people, where there are more people than dates and the pigeonhole principle takes over.

That distinction looks pedantic and is about to become the entire argument. **A rule that is right 99.99% of the time is wrong once in every ten thousand tries.** A program that

runs tens of millions of comparisons will be wrong thousands of times. Rounding that away on the page is how a probability turns into a certainty in somebody's mind. A certainty is what gets acted on.

04 It is about pairs, not people

Here is the whole mechanism. Slow down for it, because everything else in this chapter is this one idea applied to a bigger file.

A room of 20 people does not contain 20 chances for a coincidence. It contains one chance for every *pair* of people in it, and the number of pairs is 20 times 19 over two, which is 190. That is the number that matters, and it is already about half of 365.

Add one person to the room and you do not add one chance. You add one chance for every person already there. **The people grow in a line and the pairs grow in a square.** That is why the curve in Figure 1 climbs faster than intuition expects. It is the reason everything that follows behaves as it does.

People	Pairs	Chance of a shared birthday
10	45	11.7%
20	190	41.1%
23	253	50.7%
50	1,225	97.0%
100	4,950	>99.9%
366	66,795	100%

The last two rows are worth pausing on. At 100 people the chance is >99.9% – **not** 100%, however it rounds. The probability becomes exactly 1 only at 366 people, where there are more people in the room than there are dates in the year and a repeat is forced. Everywhere below that, a shared birthday is merely very likely, and “very likely” and “certain” are different claims about the world.

Now change one thing. Instead of a room, take a state's voter registration file, and instead of a birthday, take a full date of birth attached to a name.

05 What a real matching rule actually catches

New Jersey publishes its registration list county by county. Put all 21 counties together and you have **6,657,135 registration records**, of which 6,145,992 are active, each carrying a first name, a last name, a middle name and a complete date of birth.

That last field is why the exact key a cross-state matching program used can be reproduced here, on real records, rather than approximated. The Interstate Crosscheck program returned to each participating state a list of its registrations sharing **the same first name, last name and date of birth** as a registration in another participating state. Apply that key to New Jersey and the result is:

Quantity	Value
Registration records	6,657,135
Distinct first+last+DOB keys	6,653,955
Keys held by more than one registrant	3,149
Registrants sharing a key with someone else	6,329
Colliding pairs	3,212
Largest group sharing one key	4

6,329 people in one state – 0.10% of the roll – share a first name, a last name and a full date of birth with at least one other registrant. The groups are small: 3,119 of them are pairs, 29 are triples, and exactly 1 is a group of four. Nothing here is a crowd. But 3,212 is not zero either, and zero is what the purge rule assumes.

The interesting part is what happens when the key changes. The same 6,657,135 records, keyed six different ways:

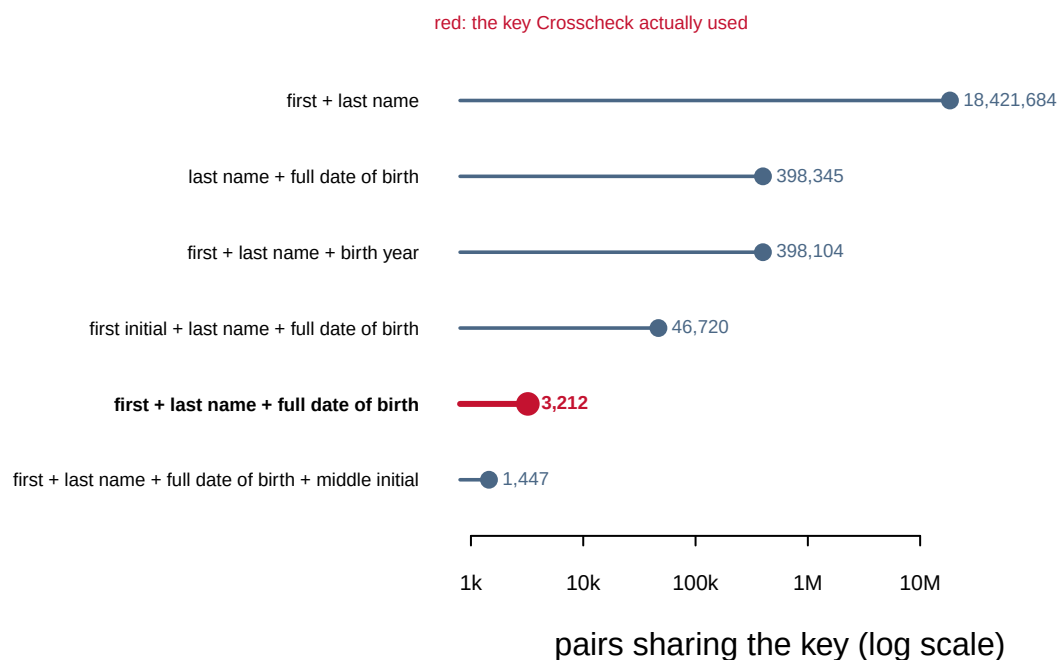


Figure 2. The same 6,657,135 records, keyed six ways. Each rung is the number of pairs of *different* registration records that a rule would treat as possibly the same person. Dropping the date of birth entirely leaves 18,421,684 pairs. Keeping only the birth year leaves 398,104. Using the full date leaves 3,212. Adding a middle initial, which these records already carry, leaves 1,447.

Three things fall out of that figure, and each of them is a decision some real election office has made.

Birth year is not date of birth. Matching on first name, last name and birth year produces 124 times as many colliding pairs as matching on the full date. That is not a hypothetical rung on the ladder. Most states, Georgia among them, publish a birth year and withhold the day and month, so anyone matching from a public file is forced onto the coarse key whether they like it or not. Houston County, Georgia has 127,560 registrants; keyed

on first name, last name and birth year, 954 of them – 0.75% – share a key with someone else.

Fuzzy matching is expensive. Abbreviating the first name to its initial, the kind of tolerance a system adds so that Robert and Bob both match, multiplies colliding pairs by 15. Loosening a key to catch more true duplicates catches disproportionately more false ones.

The cheap fix was available and unused. These records carry a middle name. Using just its first letter removes 55% of the collisions. Crosscheck's own participation guide recommended checking middle-name consistency before acting on a match; the reporting on the program suggests that recommendation was not always what member states did.

06 Two counties, one key

The 3,212 colliding pairs in New Jersey are not scattered randomly with respect to geography, and the way they are arranged is the whole problem.

Quantity	Value
colliding keys	3,149
keys spanning more than one county	2,255
keys confined to a single county	894
registrants in a cross-county key	4,531
colliding pairs	3,212
pairs that are cross-county	2,287

Of the 3,149 keys held by more than one registrant, **2,255 are held by people in different counties** – 72% of them. Only 894 are contained within a single county's own list.

This matters because it is precisely the arrangement that a single jurisdiction cannot see and a cross-jurisdiction program is built to find. A county clerk looking only at their own roll would encounter 894 of these. Pool the counties and 2,255 more appear, each one looking exactly like a person who has registered in two places.

Some of them will be. People move between counties, and a stale registration left behind is a real and common thing. **That is the difficulty, not an objection to it.** A file containing a name and a date of birth has no way to separate one person who moved from two people who never met. It does not contain the information required to tell the difference, and no amount of care applied to the matching will conjure it.

07 The rate is not a property of the rule

There is a natural way to summarize a matching rule's error – a rate, some number of false matches per hundred thousand voters, a fixed property you could look up. It does not exist, and Figure 3 is why.

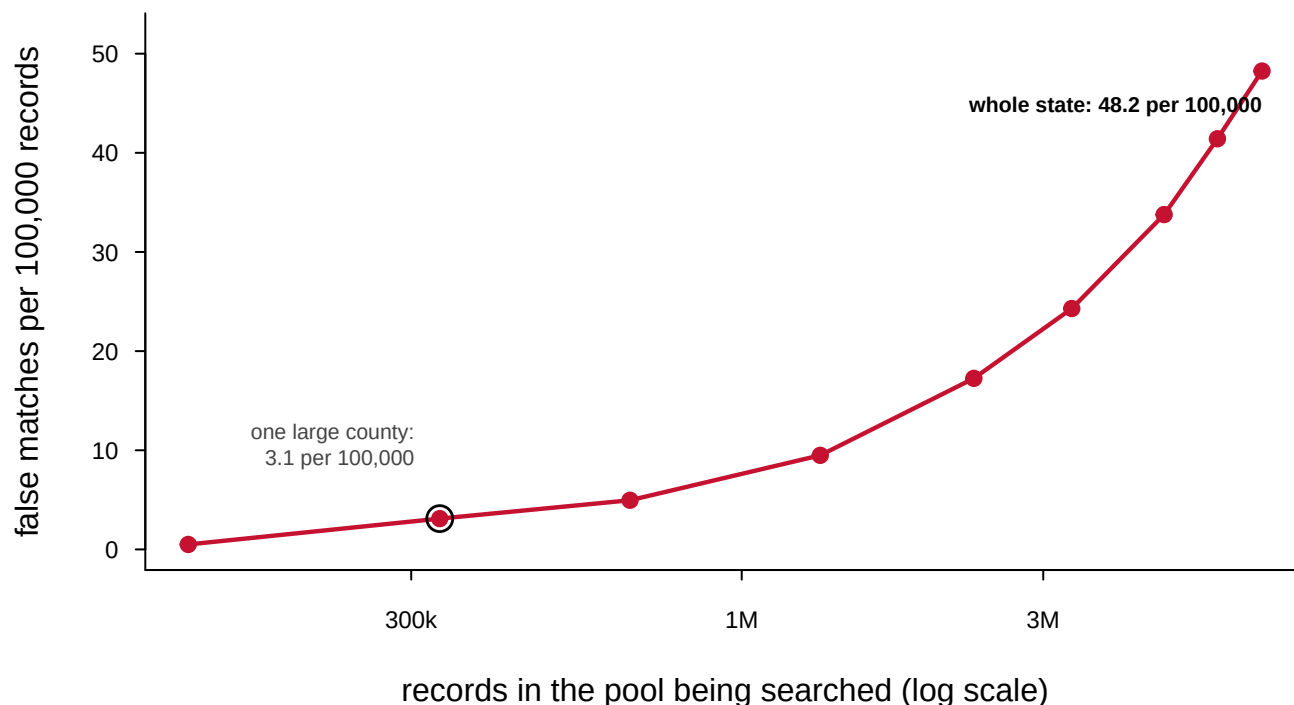


Figure 3. The same rule, the same records, a rising error rate. Each point is a random subsample of the New Jersey file at a given size, with the first-name-plus-last-name-plus-date-of-birth collisions actually counted in it. In a pool of 332,857 records the rule produces 3.1 false matches per 100,000. In the full 6,657,135 it produces 48.2 – about 16 times as many *per registrant on the roll*, from the identical rule. Both rates share a denominator, meaning the bottom number they are each divided by: records in the pool. That is the only reason they can be compared at all.

The reason is the pairs again. Doubling the file doubles the number of records that can be wrongly matched and simultaneously doubles the number of records each of them can be wrongly matched *against*. Fitting a line through Figure 3 on log axes gives a slope of 2.08, where 2 would be exactly quadratic.

A matching rule that is accurate in a county is not accurate in a state. And one that is accurate in a state is not accurate across thirty of them. The rule did not change. The pool did.

In 2012, Crosscheck ran across more than 45 million registration records.

08 The field underneath the key

One more thing sits between the rule and the truth, and it is easy to miss because it looks like a technicality.

A date of birth is only as good as whoever typed it in.

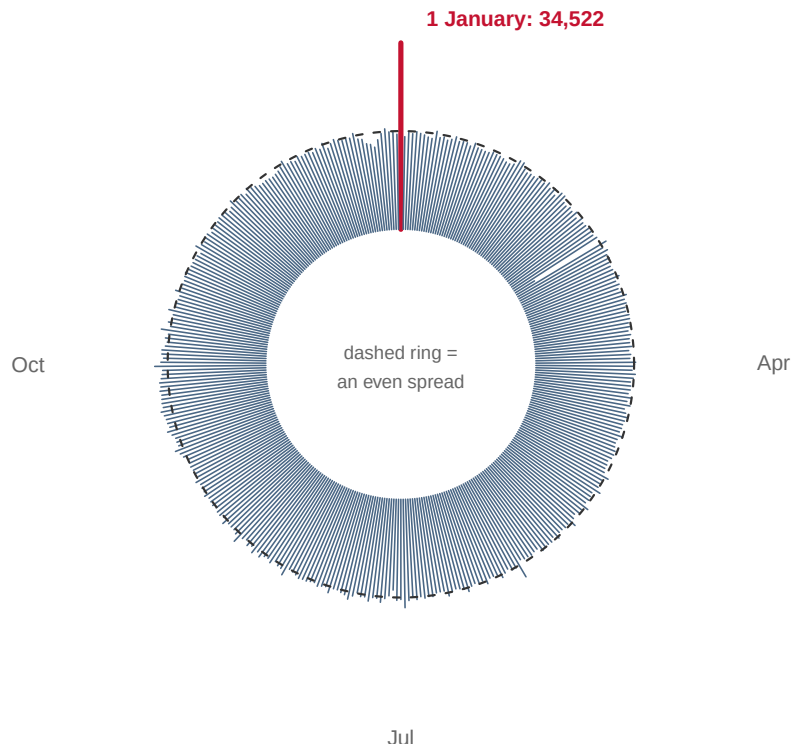


Figure 4. Every calendar date, and the number of New Jersey registrants carrying it.

One spoke per date, length proportional to count; the dashed ring is where every spoke would end if birth dates were spread evenly. 34,522 registrants have a date of birth of 1 January, which is 1.90 times the even-spread expectation. The short spoke is 29 February, which occurs in one year out of four.

The 1 January spike is not a fact about obstetrics. It is what a default value looks like: a field that had to be filled, filled with the first date available. A single sentinel date, 1 January 1800, is carried by 13,282 records in this file, and no one born in 1800 is registered to vote in New Jersey.

The honest reading of Figure 4 is narrower than it first appears, and worth stating rather than letting the picture overclaim. Clustering in a distribution can only ever *raise* the chance of a collision, never lower it. But across all 366 dates the aggregate effect here is tiny: the observed spread produces 0.5% more collisions than a perfectly even one would. Removing the 1800 sentinel entirely moves the statewide count from 3,212 pairs to 3,127.

So the spike is real, it is evidence of exactly the data-quality problem it looks like, and it is *not* by itself what drives the false-match rate. What drives the false-match rate is the coarseness of the key and the size of the pool – Figure 2 and Figure 3. A chapter that pinned the problem on dirty birthdays would be telling a more satisfying story than the numbers support.

09 The year, and the one group where the key stops working

A date has a year in it too, and the year turns out to be where the day's anomaly actually came from.

The natural place to look is 1900. It is the classic placeholder, the year a system reaches for when a field must be filled and nothing is known. In this file 1900 is almost absent: 4 records carry it. The placeholder here is **1800**, and it is not scattered across the calendar – almost every record carrying it carries 1800-01-01 exactly.

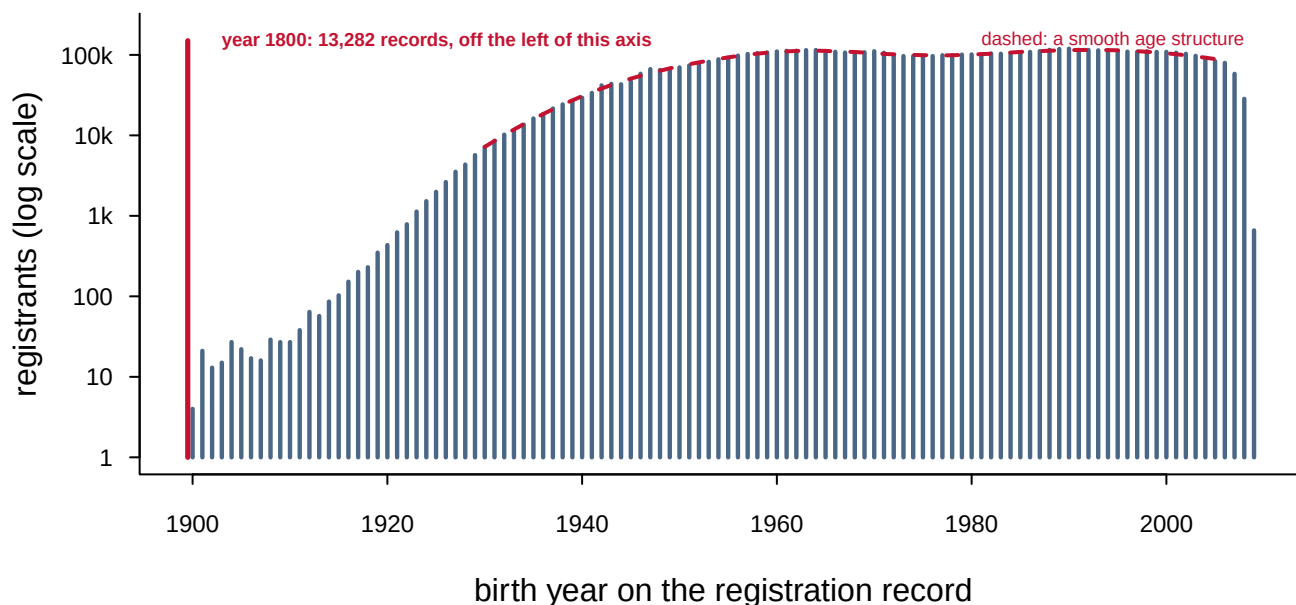


Figure 5. Birth years on 6,657,135 New Jersey registration records. One needle per year on a log scale, so a year held by one person and a year held by a hundred thousand are both visible. The dashed line is a smooth fitted through 1930 to 2005 – the null here is a plausible age structure, not a flat line, because an electorate's birth years are genuinely lumpy. The bulk follows it closely. What does not is the placeholder year 1800, drawn at the left edge because 13,282 records is off the top of any axis that also shows the single-record years beside it.

Three things are true of that figure, and only one of them is a scandal.

The age structure itself is clean. Running the standard test for data-entry rounding – comparing each decade year against the mean of its two neighbours – gives ratios between 0.97 and 1.04. There is no 1950 pile-up, no 1970 pile-up, nothing. Whatever else is wrong with this field, the years people actually typed are the years people were actually born.

There is a thin tail of plainly impossible values. Beyond the placeholder, 79 records carry a birth year before 1900 – three-digit years, a 1652, a 1776. Set the placeholder aside and 7,980 registrants have an implied age over 100 and 566 over 110.

Some of those are real centenarians, and New Jersey has them. Others are records of people who have died and have not been removed.

That distinction matters, because removing dead registrants is precisely what list-maintenance programs exist to do – and nothing in a name and a date of birth identifies

which is which. At the other edge, 659 records imply an age under 18, which is not an error: New Jersey lets 17-year-olds pre-register.

And the placeholder is where the matching rule quietly stops working. This is the part that belongs in this chapter rather than in a note about data hygiene. Inside the group of 13,282 records sharing 1800-01-01, the date of birth is the same for everybody. It therefore contributes nothing to the key. First name plus last name plus date of birth **collapses to first name plus last name** – the top rung of Figure 2, the rung with 18,421,684 colliding pairs.

The consequence is measurable:

Quantity	Value
Records carrying the placeholder date	13,282
...as a share of the file	0.20%
Colliding pairs inside that group	85
...as a share of all colliding pairs statewide	2.6%
Colliding pairs in 20 same-sized draws of real dates (mean)	0.0
...the worst of those 20 draws	0

0.20% of the records produce 2.6% of the collisions – an over-representation of about 13 to one, where both shares are taken over the same statewide file of 6,657,135 records. And the comparison group is the striking part: taking 13,282 records with genuine dates of birth, twenty times over, produced 0.0 colliding pairs on average and never more than 0. Against 85.

A matching rule fails worst on the records that were already the most damaged. Some registrants had their date of birth recorded badly or not at all. They are disproportionately old registrations, records transferred between systems, or people who registered somewhere that did not capture the field. Those are the ones the rule is most likely to flag as somebody else.

That also settles the loose end left by Figure 4. The 1 January spike looked like a general problem with birthdays. It is largely this one group.

Among registrants born from 1930 onwards, 1 January is over-represented by a factor of 1.17, against 1.90 for the file as a whole.

10 The same defect, four times worse, in the file that mattered

None of this is a peculiarity of New Jersey, and it is not a new discovery. It is documented in the peer-reviewed literature, at national scale, and it was severe enough to change what a published paper could estimate.

When **Goel and colleagues (2020)** assembled a national file of 2012 vote records, they found that **14% of them carried a first-of-the-month birthday**. Their word for that is *improbable*, and it is the right one. Twelve of 366 dates should account for about 3.3% of records, not one in seven. McDonald (2007) had observed that election officials sometimes use a first-of-the-month date to mark missing information. Following that, they

treated those records as carrying stand-in dates of birth and **deleted them before estimating anything**. Leaving them in would inflate the count of apparent double votes.

The comparison with New Jersey is worth making carefully, because the two numbers are not alike and the difference is the interesting part.

File	First-of-the-month share
New Jersey registration records, this chapter	3.71%
National 2012 vote records, Goel et al.	14%
If dates of birth were spread evenly	3.28%

New Jersey's own extract is **far cleaner** – an excess of only 1.13 times, most of it the single placeholder date already discussed. The national file the researchers had to work with was four times worse.

That gap is the argument for reading this chapter's New Jersey numbers as a floor rather than a typical case. A state's own extract is one office's record-keeping. A national file is many offices' stitched together.

A pooled matching program does not inherit the average quality of the files it searches. It inherits the problems of every one of them at once. Crosscheck pooled 15 states. The collisions counted in Figure 2 were found in the cleanest kind of file this analysis could have used.

11 An estimate that was already wrong, about this state

There is a precedent for all of this, and it is on New Jersey.

In 2008, McDonald and Levitt published what was then the most extensive study of double voting yet attempted, examining it within New Jersey in the 2004 presidential election. They found **884 pairs** of vote records sharing a first name, last name and date of birth – the same key this chapter has been applying, on the same state's rolls, two decades earlier.

The question they then faced is the one this whole chapter is about: how many of those pairs are two different people? They simulated coincidental matches, drawing each record's birth year from the real age distribution of voters and **assuming that birthdays within a year follow a uniform distribution**. That produced a 95% confidence interval of **300 to 500 people voting twice in New Jersey** in 2004.

Goel and colleagues (2020) revisited that estimate with two corrections. They allowed for the clustering in names and days of birth measured in Figures 4 and 5. They also allowed for the possibility that a registration record is wrongly marked as having been used to vote. Their conclusion is that the true number of double votes is **fewer than one-tenth of what the earlier approach suggests**.

That is this chapter's entire argument, already played out once, on this state's data. An assumption of uniform birthdays sounds harmless, sounds even conservative, and inflated a published estimate of electoral fraud by a factor of ten.

Be exact about why, because the reason is not the one the phrase "birthdays are not uniform" suggests. Figure 4 showed that the *smooth* part of the variation – seasons, days of

the week – barely matters: across all 366 dates the aggregate effect on collisions here is 0.5%. What breaks the uniform assumption is not that real birthdays are lumpy. It is that a substantial block of recorded birthdays **are not birthdays at all** – they are placeholders, 14% of them in the national file, piled onto twelve dates. Assume those are uniformly distributed real birth dates and the model will conclude that a great many people improbably share one, which is exactly what “improbable coincidence” is taken to mean.

One more sentence in that paper does more damage to the fraud narrative than any collision count above. Even taking the authors’ own headline estimate at face value, they note something striking. A clerical error rate of merely **1.3%**, meaning records wrongly marked as having voted, would be **enough to explain all of the apparent double votes they found**.

12 What was done with the matches

Between roughly 2005 and 2019, the Interstate Voter Registration Crosscheck program pooled registration records from participating states. It returned to each state a list of its own registrations that shared a first name, last name and date of birth with a registration somewhere else. In 2012 it processed more than 45 million records. In 2013 it circulated a document reporting that it had identified **1,395,074 “potential duplicate voters”** among the 15 states then participating.

What that number means was the subject of a study by Sharad Goel, Marc Meredith, Michael Morse, David Rothschild and Houshmand Shirani-Mehr, published in the *American Political Science Review* in 2020. They filed public-records requests with every state then in the program. **Iowa was the only one able to supply the data in usable form**. What it sent was the list of 100,140 pairings Crosscheck had provided to the Iowa Secretary of State before the 2012 election.

Crucially, those records carried something no public voter file does. They said whether the last four digits of each registrant’s Social Security number matched. That is the field that can tell one person from two. It is why every figure in this section rests on one state’s copy of a multi-state program’s output.

Their findings are more interesting than the obvious guess, and they cut in two directions at once.

Finding	Value
Iowa pairings Crosscheck flagged, SSN4 known for both records	34,900
of those Iowa pairings, the share that really were one person	70-75%
of those confirmed duplicates, the number used to vote twice in 2012	fewer than 10
of those Iowa pairings, pairs where both voted and the SSN4 matched	7
of those Iowa pairings, pairs where both voted and the SSN4 differed	1,476
double voters per voter nationally in 2012, all US voters as the base	one in 4,000

The last row is a different kind of quantity from the five above it, which is why it says so. It is not a rate within Crosscheck’s matches. The abstract’s wording is exact and worth keeping. “We estimate that about one in 4,000 voters cast two ballots.”

The derivation is 32,890 double voters among the roughly 129 million people who voted in the 2012 presidential election nationally. Read instead as a share of the flagged pairs,

it would mean something else entirely. The paper itself phrases it both ways in different places, which is precisely why the base is spelled out here.

Most flagged registrations really were duplicates. Among the Iowa pairings where the Social Security digits were known for both records, roughly three-quarters were genuinely the same person registered in two states. The match was not usually wrong about that. So the intuitive complaint – that name-and-birthday matching mostly produces nonsense – is not what the data show, and you should drop it.

And almost none of them voted twice. Of the 25,987 confirmed duplicate registrations, fewer than 10 were used to cast a ballot twice. Being registered in two states is what happens when you move and do not file paperwork you had no reason to know existed. It is not fraud, and it is not evidence of fraud.

Where the matches did mislead was on the question that mattered. Look only at pairs where both registrations were actually used to vote, which are the ones that look like double voting. The Social Security digits matched in 7 cases and differed in 1,476.

The authors measure that three ways, and the spread is worth seeing. In the national voter file, 97% of votes cast under the same first name, last name and date of birth came from two different people. Restricted to Crosscheck states, 99.4%. Measured directly in the consortium data, where the Social Security digits settle it, 99.5%. Three methods, three files, one answer. **A match on those three fields is overwhelmingly two people.**

Put those together and you get the finding that matters for policy. Several states, Indiana among them, adopted or considered a rule cancelling the older of two registrations sharing a name and date of birth. Applied to the consortium states' own data, the authors estimate that such a rule would **impede approximately 300 legitimate votes for every double vote prevented.**

That ratio is between two different things and both halves are worth stating plainly, because a bare "300 to 1" invites exactly the question this chapter keeps asking. The numerator, the top number of the fraction, is registrations that were used to cast an apparently legitimate vote and that the rule would cancel: more than 2,500 of them in the consortium data. The denominator is double votes actually prevented, of which there were fewer than 10 to prevent among roughly 26,000 confirmed duplicate registrations. It is not a share of anything; it is a price, quoted in lost legitimate votes per fraudulent one stopped.

The program was suspended in December 2019 as part of a settlement of a lawsuit brought by the ACLU of Kansas over its handling of voter data, and it has not run since.

13 What the arithmetic will and will not carry

The chain in this chapter is short and each link is a different kind of claim, so separate them before anyone repeats the conclusion.

Two links are direct measurement. The birthday curve in Figure 1 is arithmetic with no data in it at all and no assumptions to argue with. The collision counts in Figures 2 and 3 are counts of real registration records. 6,657,135 of them across 21 New Jersey counties, plus 127,560 in Houston County, Georgia. All keyed and tallied, with no model in between. The scaling curve is measured by subsampling the real file rather than projected from a formula.

One link is borrowed, and it is the one carrying the most weight. Nothing in a public voter file can establish whether a matched pair is one person or two. That determination required the Social Security digits. It comes entirely from Goel and colleagues, working from the one state, Iowa, that could supply Crosscheck's output with those digits attached. This chapter counts the matches. It does not and cannot verify them.

The same applies to the New Jersey precedent above. McDonald and Levitt's 884 pairs, and the finding that the true figure is fewer than one-tenth of their estimate, are reported here as Goel and colleagues describe them, rather than re-derived.

And several things here are genuinely limited. New Jersey is one state, and the cross-county comparison inside it is a stand-in for a cross-state comparison rather than the thing itself. Counties within a state share migration patterns and naming distributions that thirty scattered states do not.

Georgia's contribution is a single county. The 124-fold penalty for using birth year instead of the full date is measured on New Jersey's records. It would differ elsewhere. No number in this chapter should be read as a national false-positive rate, and none is offered as one. That is exactly why Figure 3 shows the growth curve instead of a single headline figure.

What survives all of that is a claim about inference rather than about arithmetic. A name and a date of birth are not an identifier.

They were never designed to be one. They are not unique in a population of any size. And the number of ways they can coincide grows with the square of the number of people you check. Treating a match on those fields as proof of identity is a decision to accept a certain number of wrongly flagged voters. The only real question is whether whoever adopts the rule knows what that number is, and says so.

6,329 people in New Jersey share a name and a birth date with a stranger. Almost all of them will never know it. Whether that fact costs any of them a vote depends entirely on what somebody decides a match means.

14 Sources

Data

- New Jersey statewide voter registration extract, `vlist_<County>.csv`, all 21 counties, consulted 11 August 2026. 6,657,135 registration records carrying first name, last name, middle name, suffix and full date of birth. Public records, but not posted for download: the Division of Elections releases the list on request, which is how this copy was obtained. <https://www.nj.gov/state/elections/index.shtml>
- Georgia Secretary of State, Houston County voter registration extract, consulted 11 August 2026. 127,560 registration records carrying first name, last name, middle name and **birth year only**. Public records. Georgia sells the statewide list to anyone who orders it; this copy arrived as working material in a county redistricting matter. <https://sos.ga.gov/page/order-voter-registration-lists-and-files>
- Neither file is redistributed with this chapter, and no name, date of birth, address or registration number from either appears anywhere above. Every figure and table is built from counts alone.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`birthday.csv`, `birthyears.csv`, `crosscounty.csv`, `ga_keys.csv`, `groupsize.csv`, `keys.csv`, `monthday.csv`, `scaling.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `keys.csv` and `ga_keys.csv` show what happens to a match rule as you add or remove parts of the key. Compare the pct flagged. Which single addition does the most work?
- `groupsize.csv` counts how many registrants share a key, by group size. Find the largest group. How many unrelated people would a purge rule have treated as one person?
- `monthday.csv` counts registrations by birth month and day. Look at the first of January and the first of the month generally. What is going on, and what does it do to a matching rule?
- `scaling.csv` shows collisions rising with the number of records. Follow `per100k`. If a state doubled its roll, what happens to the false matches?

Academic literature

- Goel, S., Meredith, M., Morse, M., Rothschild, D. and Shirani-Mehr, H. (2020). "One Person, One Vote: Estimating the Prevalence of Double Voting in U.S. Presidential Elections." *American Political Science Review* 114(2): 456–469. doi:10.1017/S000305541900087X — **every claim above about Crosscheck's matches, the Social Security comparison, the 14% first-of-the-month finding and the approximately 300-to-one ratio is theirs.** Page citations throughout are to the published article, checked against the final text; `facts.csv` records the page and the quoted sentence behind every one of them.
- McDonald, M. P. and Levitt, J. (2008), on double voting in New Jersey in 2004, and McDonald, M. P. (2007), on placeholder dates of birth — both discussed above **as Goel et al. report them** (pp. 457, 459) rather than read directly, which is why no figure is attributed to them here beyond what that article states.
- The Indiana provision described above is Ind. Code Ann. 3-7-38.2-5(d)(2); Goel et al. cite it alongside *Common Cause Indiana v. Lawson*, 937 F.3d 944 (7th Cir. 2019).

How the figures were made

Both registration files are read from their original locations, with their row counts and the date they were consulted recorded. They are reduced to totals only. Collision counts at six key precisions, the cross-county breakdown, the subsample scaling curve, the calendar and birth-year distributions, the placeholder-group comparison behind Figure 5, and the exact birthday probabilities. Every number printed above is computed from those files at the moment this page was built. None is typed into the prose.

Two conventions in that script are worth naming because they are the kind of thing that quietly goes wrong. Percentages are formatted by a single function that refuses to print a probability below 1 as "100%". The birthday probability rounds to exactly 1.0 in the computer's arithmetic at 153 people, while remaining below 1 mathematically until 366. So certainty is read off the number of people, never off the stored value. And every figure is drawn twice, once for the browser and once for print, from the same files rather than from each other.